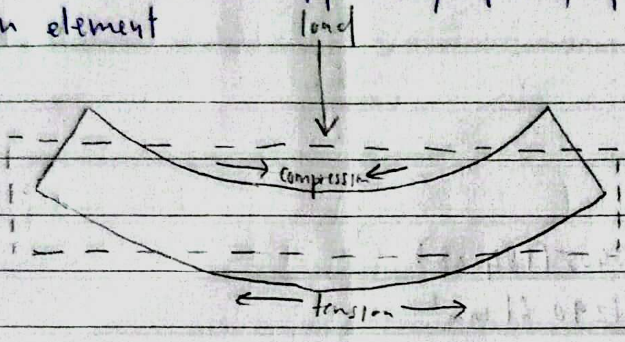
No.: Date:

SPM II 20~~24~~/2025

1. a) - ~~Bending~~ the behaviour of a slender structural element subjected to an external load applied ~~perpend.~~ perpendicular to an axis of an element



b)

characteristic of steel

in elastic region	in plastic region
- Returns to original shape	- Permanent change
- No damage since the steel is not permanently bent	- Yielding since the steel is passes it's "limit" (Yield point)
- Linear, the more forces is added the more it stretches in perfect linear line ratio	- Ductility, the steel acts like plasticine.

importance of yield point

ii) - safety limit, it marks the maximum load a material can handle before it is permanently ~~chang~~ damaged.
- Warning sign, it allows the steel to bend and ~~streth~~ stretch before snapping.

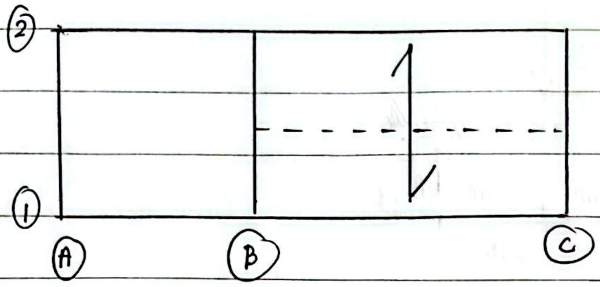
c)

environmental factor	The effect
Wind	- the tall buildings will sway - caused by moving air hitting the tall building's surface
seismic forces	- the tall building will get shaken - caused by ground motion acting on the tall building's mass.

CamScanner

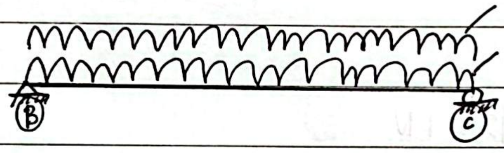
2. (a)

(i)



$\frac{l_y}{l_x} = \frac{8500\ mm}{4000\ mm} = 2.125 > 2$; one-way slab

(ii)



(iii) s/w of beam $= 25(0.2)(0.35-0.15)$

$= 1\ kN/m$

(iv) s/w of slab $= 25(0.15)(2.0)$

$= 7.5\ kN/m$

(v) Total G_k of slab = s/w of slab + floor finishing

$= 7.5\ kN/m \quad 1.5\ kN/m^2 \times 2m$

$= 7.5 + 3$

$= 10.5\ kN/m$

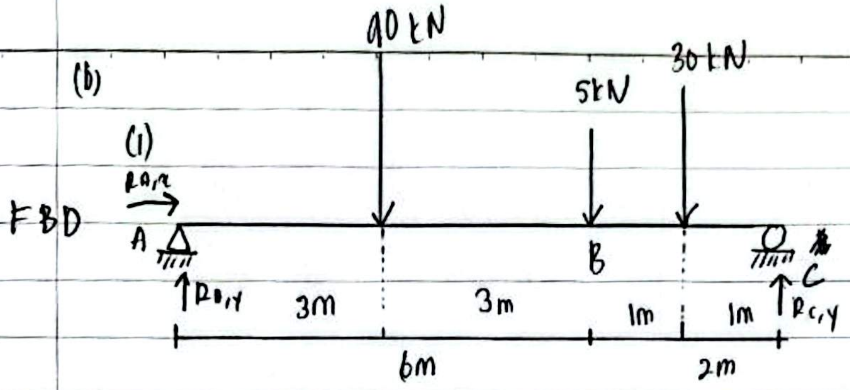
(vi) Ultimate load $= 1.35G_k + 1.5Q_k$

~~$1.35(10.5+7) + 1.5(2.0)$~~

$= 1.35(10.5+1) + 1.5(2.0 \times 2)$

$= 21.525\ kN/m$

(b)



(II) $\overset{+}{\rightarrow} \Sigma F_x = 0$

$R_{A,x} = 0$ kN

$+\uparrow \Sigma F_y = 0$

$R_{A,y} - 90 - 5 - 30 + R_{C,y} = 0$

$R_{A,y} + R_{C,y} = 125$ kN

$\circlearrowleft + \; M_A = 0$

$90(3) + 5(6) + 30(7) - R_{C,y}(8) = 0$

$R_{C,y} = 63.75$ kN

$R_{A,y} + 63.75 = 125$

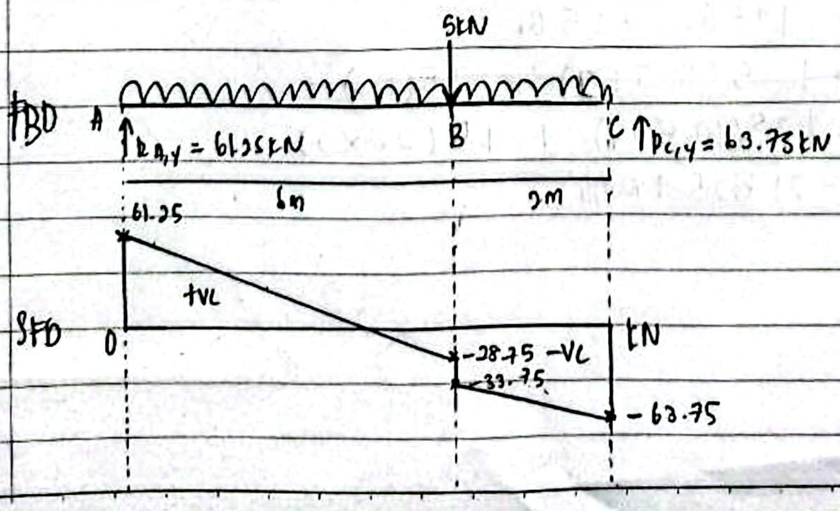
$R_{A,y} = 61.25$ kN

(III) Shear forces

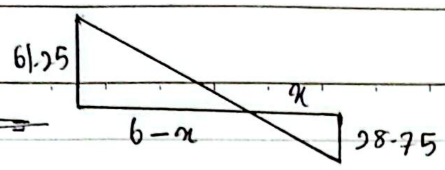
$V_A = 61.25$ kN

$V_B = \cancel{90}\ 61.25 - 90 - 5 = -33.75$ kN

$V_C = -33.75 - 30 = -63.75$ kN



(iv) ~~M max =~~



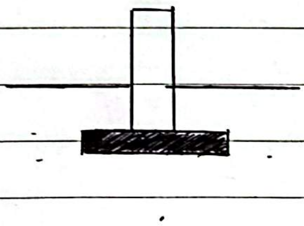
$$\frac{61.25}{6-x} = \frac{28.75}{x}$$

$$x = 1.92 \text{ m}$$

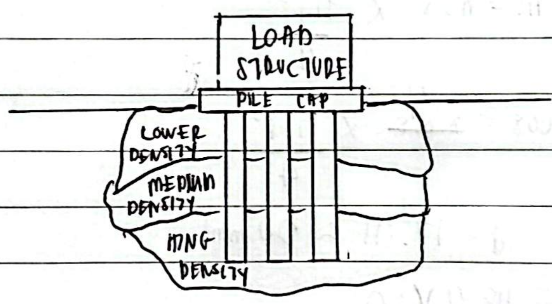
$$M_{max} = \frac{1}{2}(61.25)(6-1.92)$$
$$= 124.95 \text{ kNm}$$

3. ~~(a)~~ (a)

Shallow Foundation



Deep Foundation



Shallow Fondation	Deep Fondation
~~For~~ Foundations are constructed near to the finished ground surface	Foundation are constructed too deeply below the finished ground surface
Founding depth < 3m	Founding depth > 3m
Foundation for light structures	Foundation for heavy /tall structures

(b) Service load $= 1.0\,G_k + 1.0\,Q_k$
$= 1.0\,(700) + 1.0\,(400)$
$= 1100$ kN

Design load $= 1.35\,G_k + 1.5\,Q_k$
$= 1.35\,(700) + 1.5\,(400)$
$= 1545$ kN

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(ii) $\frac{A_s f_{yk}}{bh f_{a}} = 0.09$

$f_{yk} = 500\ N/mm^2$
$f_{a} = 35\ N/mm^2$

$\frac{A_s (500)}{(400)(400)(35)} = 0.09$

$A_s = 1008\ mm^2$

(iii) nos. $= 4$

$A_s = \text{nos} \times \frac{\pi d^2}{4}$

$1008 = \cancel{\text{nos}}\ 4 \times \frac{\pi d^2}{4}$

$d = 17.91 \approx 20\ mm$

$\therefore$ use $4\ Y20$

(iv)

ϕ link: $\frac{1}{4} \times 20 = 5mm$; $6\ mm$ ✓

link spacing: $20 \times 20 = 400\ mm$; $400\ mm$; $400\ mm$ ✓

$\therefore S_{max} = 400\ mm$

Use R6 – 400 c/c

(v) $q = 200\ kN/m^2$

$P = 1.0\,G_k + 1.0\,Q_k$

$= 1.0(15 + 700) + 1.0(400)$

$= 1115\ kN$

$q = \frac{P}{A}$

$200 = \frac{1115}{A}$

$A = 5.575\ m^2$

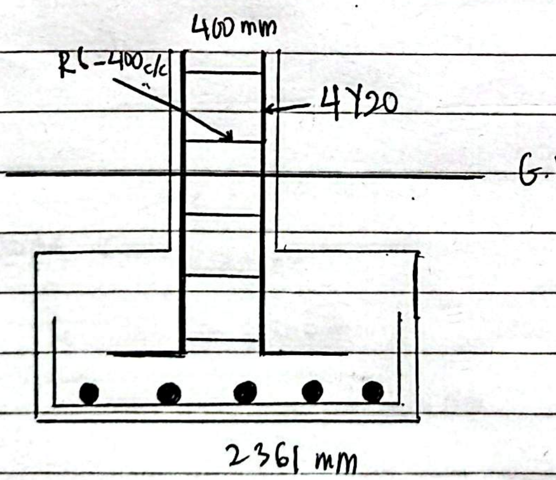
$L = \sqrt{5.575}$

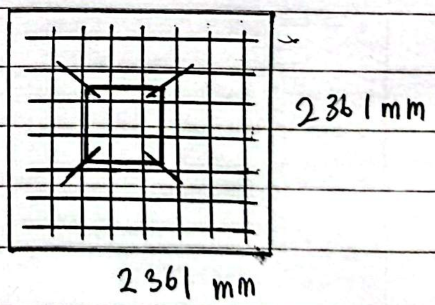
$= 2.361\ m$

$= 2361\ mm$

size of pad footing $= 2361\ mm \times 2361\ mm$

(vi)

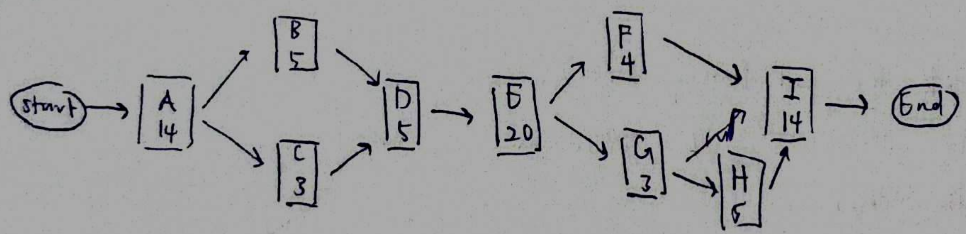


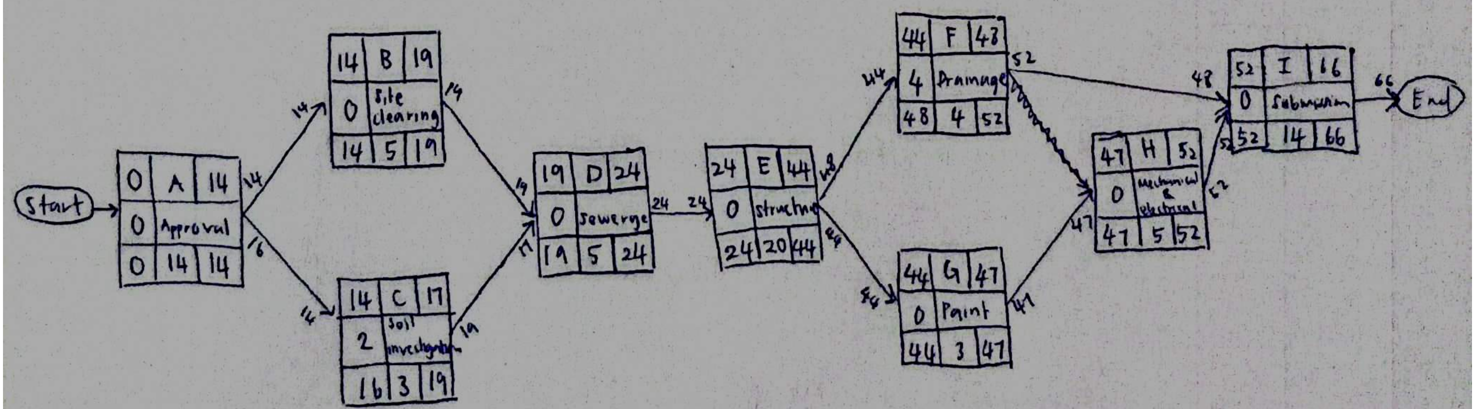


4. a)

Project life cycle	Explanation
Project initiation	- involves starting up the project. by documenting a business case, feasibility study, terms of reference, appointing the team and setting up a Project Office.
Project Planning	- involves setting out the road map for the project by creating the following plans: project plan, resource plan, financial plan, quality plan, acceptance plan and communication plan.
Project Execution	- Involves building the deliverables and controlling the project delivery, scope, costs, quality, risks and issues.
Project Closure	- involves releasing the final deliverables to the customer, handing over project, terminating supplier contracts, releasing project resources and communicating project closure to all stakeholders

b) i)





Pathway, 1: start → A → B → D → E → F → I → End = 14+5+5+20+4+14 = 62 days

2: start → A → C → D → E → G → H → I → End = 14+3+5+20+3+5+14 = 64 days

ii) Critical path: start → A → C → D → E → G → H → I → End

minimum duration: 66 days

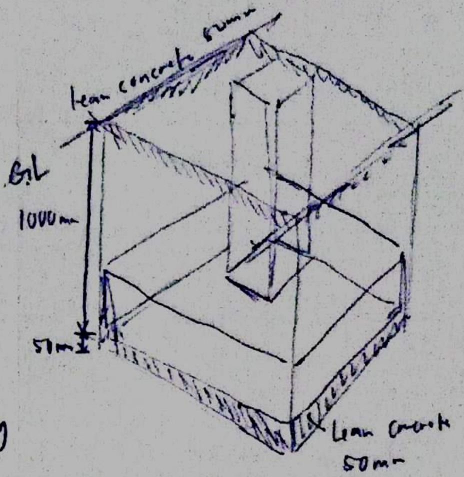
c)

i) Excavation for pad footing (m^3)

Length = 1200mm

Width = 1200mm

Depth = 1000 + 400 + 50 = 1450 mm

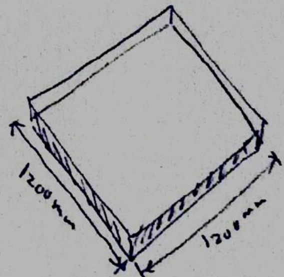


1.20 1.20 1.45	2.09	Excavation for pad footing beginning from prepared level; not exceeding 1.5m deep; backfilling and compacting the soil; transporting and dumping surplus soil to an ~~are~~ averge distance of 50m from the site.

ii) Lean concrete for pad footing (m^2)

Length = 1200mm

Width = 1200 mm

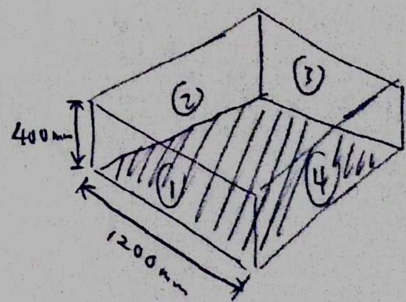


1.20 1.20	1.44	1:2:4 concrete mix for pad footing lean concrete

iii) Formwork for pad footing (m^2)

Length = 1200 mm × 4 = 4800 mm

Depth = 400 mm

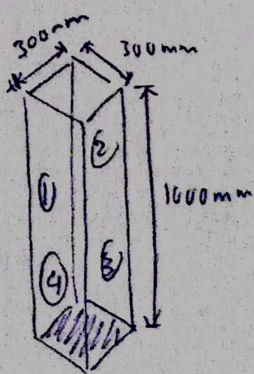


4.80 0.40	1.92	Forwork for pad footing

iv) Formwork for stump (cm^2)

Heigth = 1000mm

Length = 300 mm × 4 = 1200mm



1.00 1.20	1.2	Formwork for stump